

Mistral AB-UPT

Anchored-Branched Universal Physics Transformers

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How it works

01 Encode geometry

Pool local geometry into tokens and exchange global information.

02 Couple surface and volume

Interacting Transformer branches learn the flow around the car.

03 Predict at query positions

Anchor attention evaluates fields on large output meshes.



AB-UPT paper

arxiv.org/abs/2502.09692



Mistral AI

mistral.ai

Submission setup

400

training cases

34

validation cases

50

test cases

Trained from scratch

500 epochs

65k geometry points

16k surface/volume points

Surface + volume outputs

Forces integrated from fields

15 h

training

30 s

inference

Each on 1 H100 GPU. Inference: full-resolution .vtu.

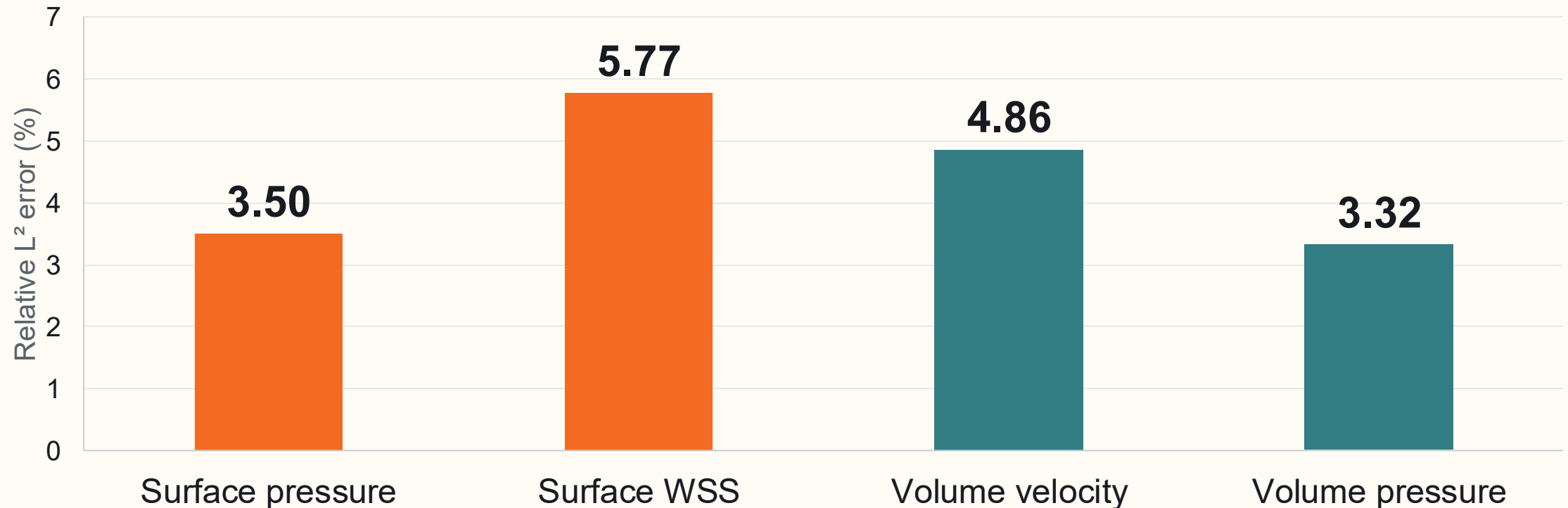
Forces and geometry-to-geometry changes

Metric	Cd	CL	Cm
R ²	0.9856	0.9859	0.9963
RMSE (counts)	3.59	6.78	2.51
Delta RMSE (counts)	4.34	9.14	3.59
Correct direction (%)	96.90	96.65	98.53

Surface and volume field errors

Overall Score: 85

Mean relative L^2 error across the 50 Full-split test cases



Surface: area-weighted cells. Volume: equal cell weights.

Velocity profiles at four stations

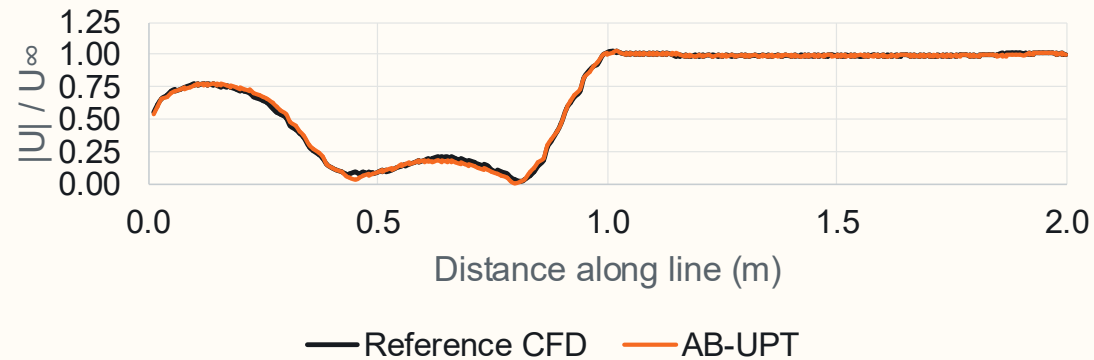
Typical case · Run 215 · V3–V6, $y = 0$ m

Overall velocity-profile R^2

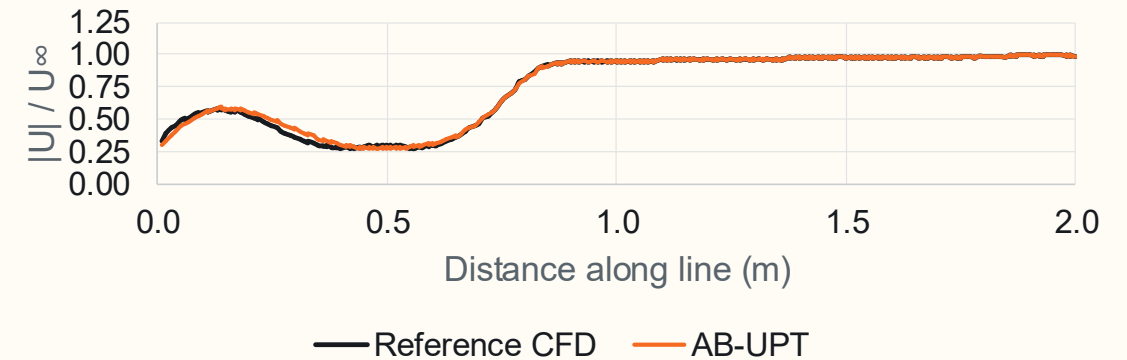
0.9966

16 lines × 50 test geometries

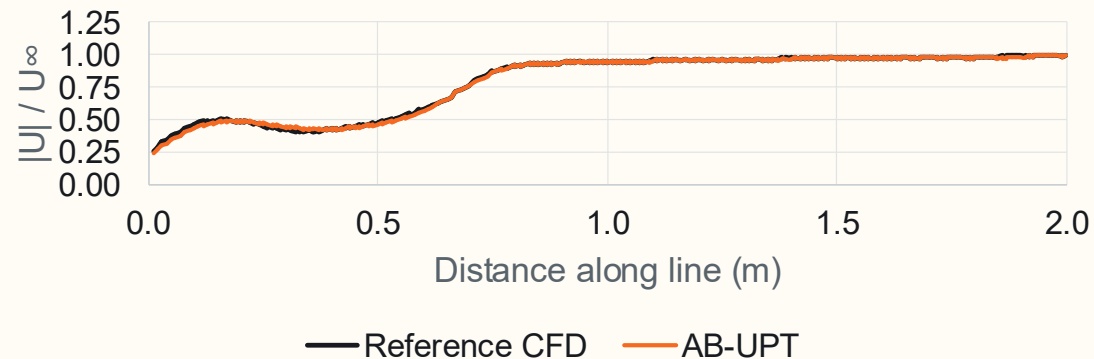
V3 · $x = 4.007$ m



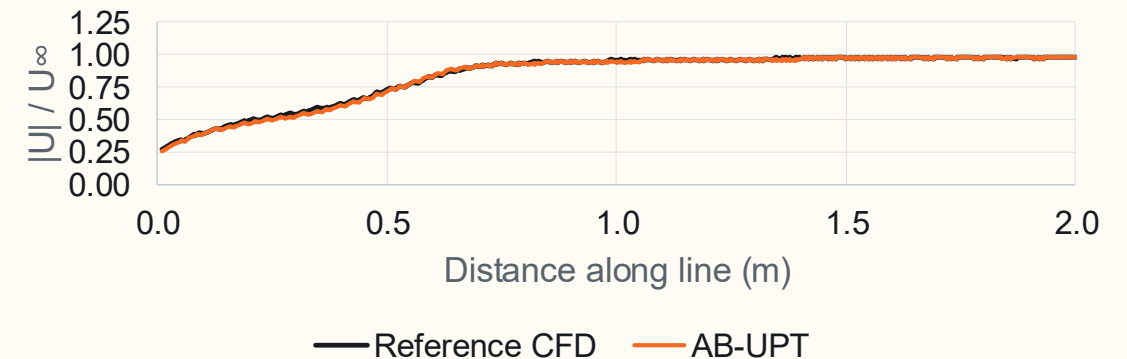
V4 · $x = 4.657$ m



V5 · $x = 5.007$ m



V6 · $x = 5.507$ m



Velocity profiles across four U cuts

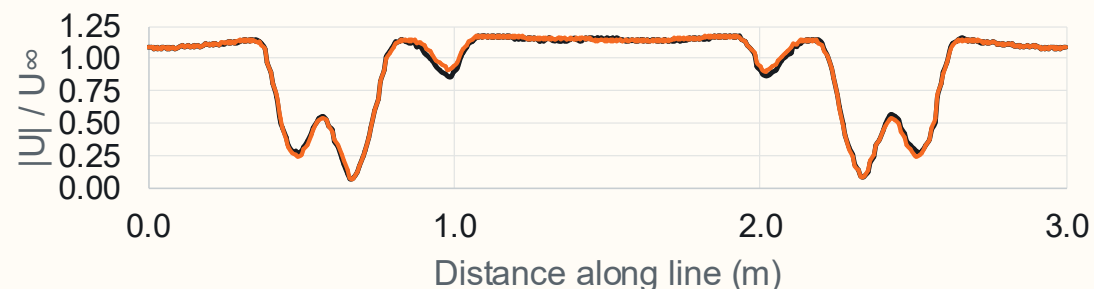
Typical case · Run 215 · U2–U5, $z = -0.2376$ m

Overall velocity-profile R^2

0.9966

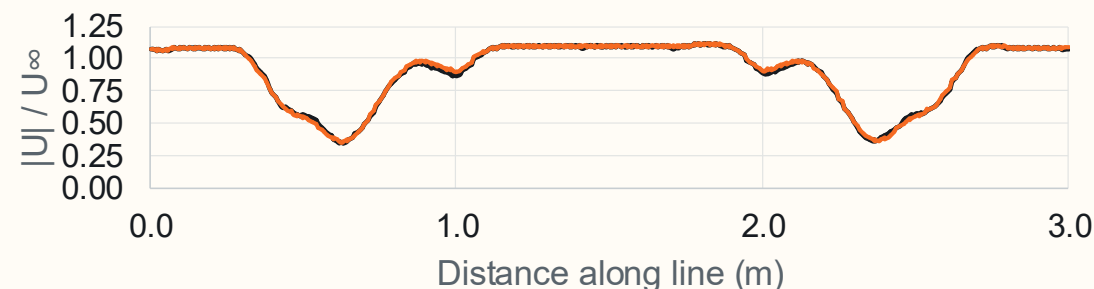
16 lines × 50 test geometries

U2 · $x = 0.357$ m



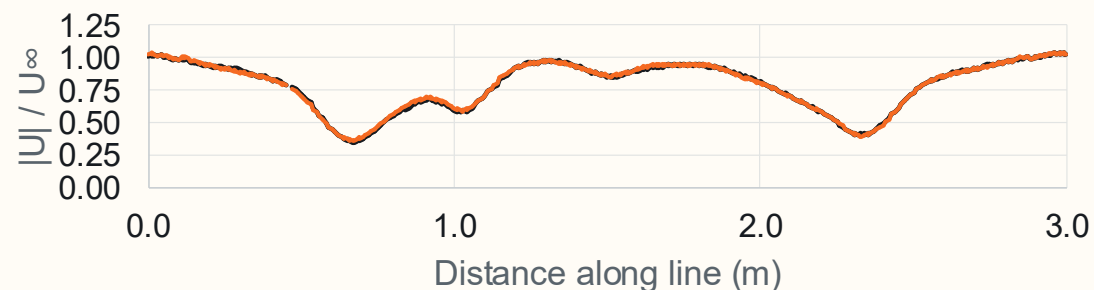
— Reference CFD — AB-UPT

U3 · $x = 0.707$ m



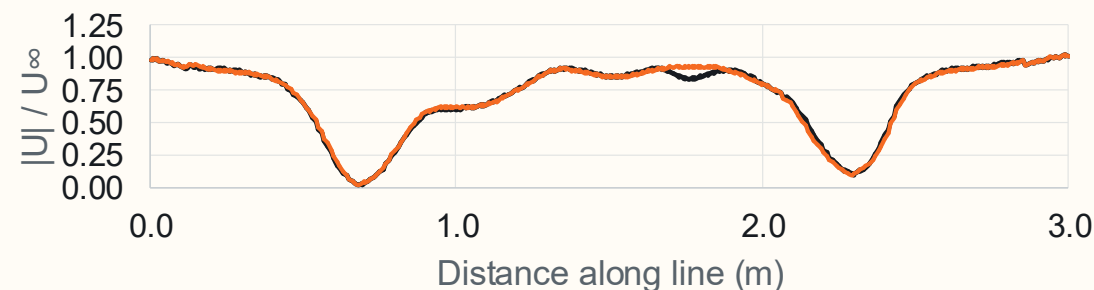
— Reference CFD — AB-UPT

U4 · $x = 2.457$ m



— Reference CFD — AB-UPT

U5 · $x = 3.157$ m



— Reference CFD — AB-UPT

Pressure cuts through the centre plane

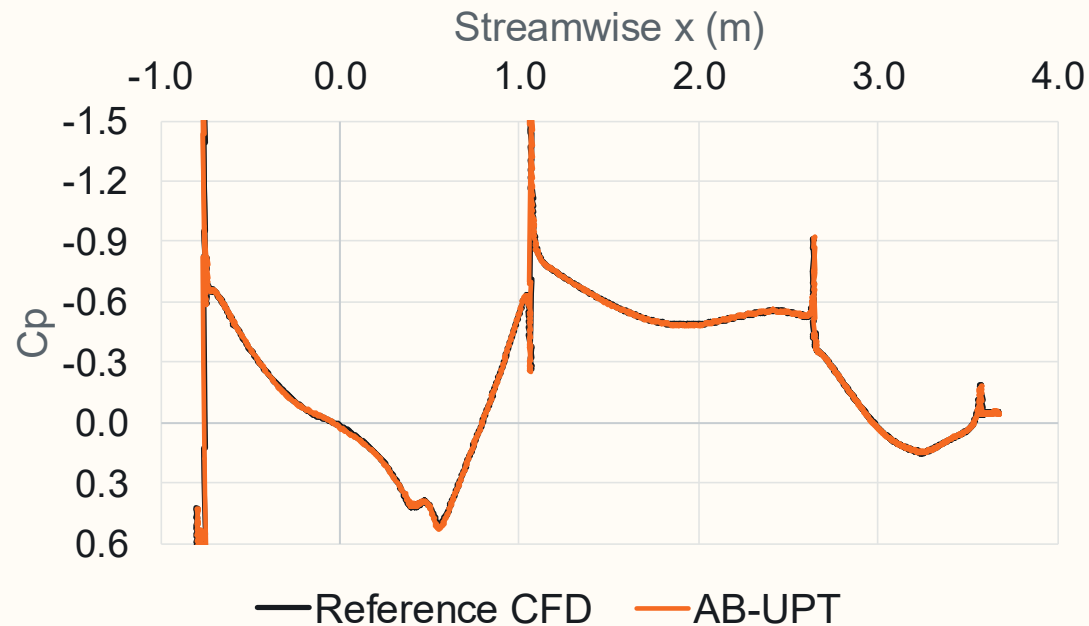
Run 215 · $y = 0$ m · upper-body and underbody surfaces

Overall Cp-profile R^2

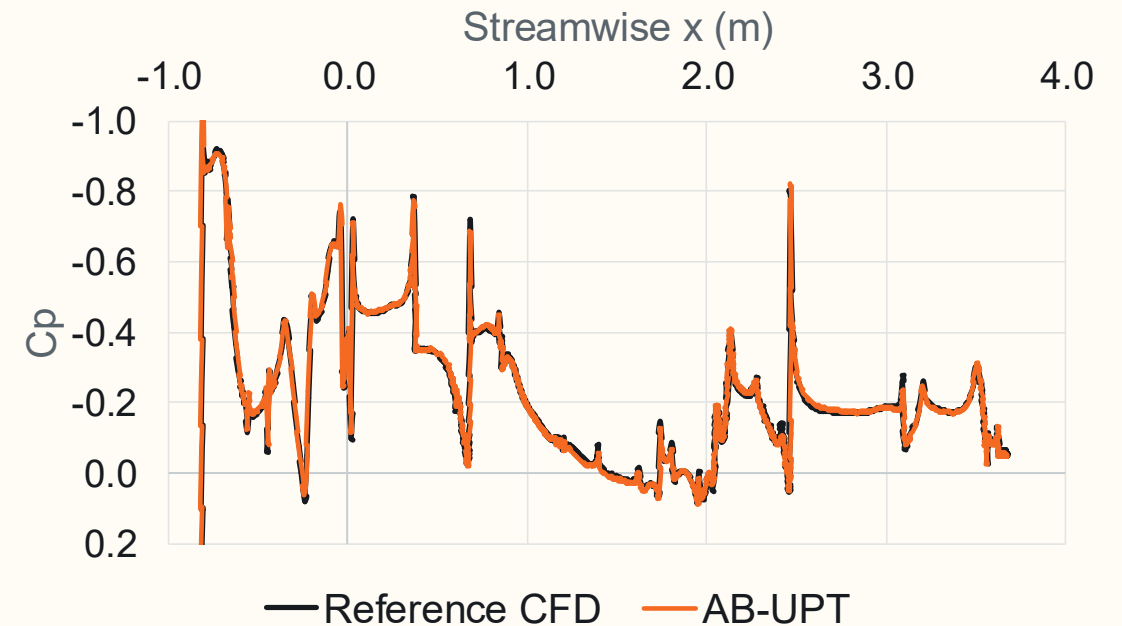
0.9985

Full split · 50 test geometries

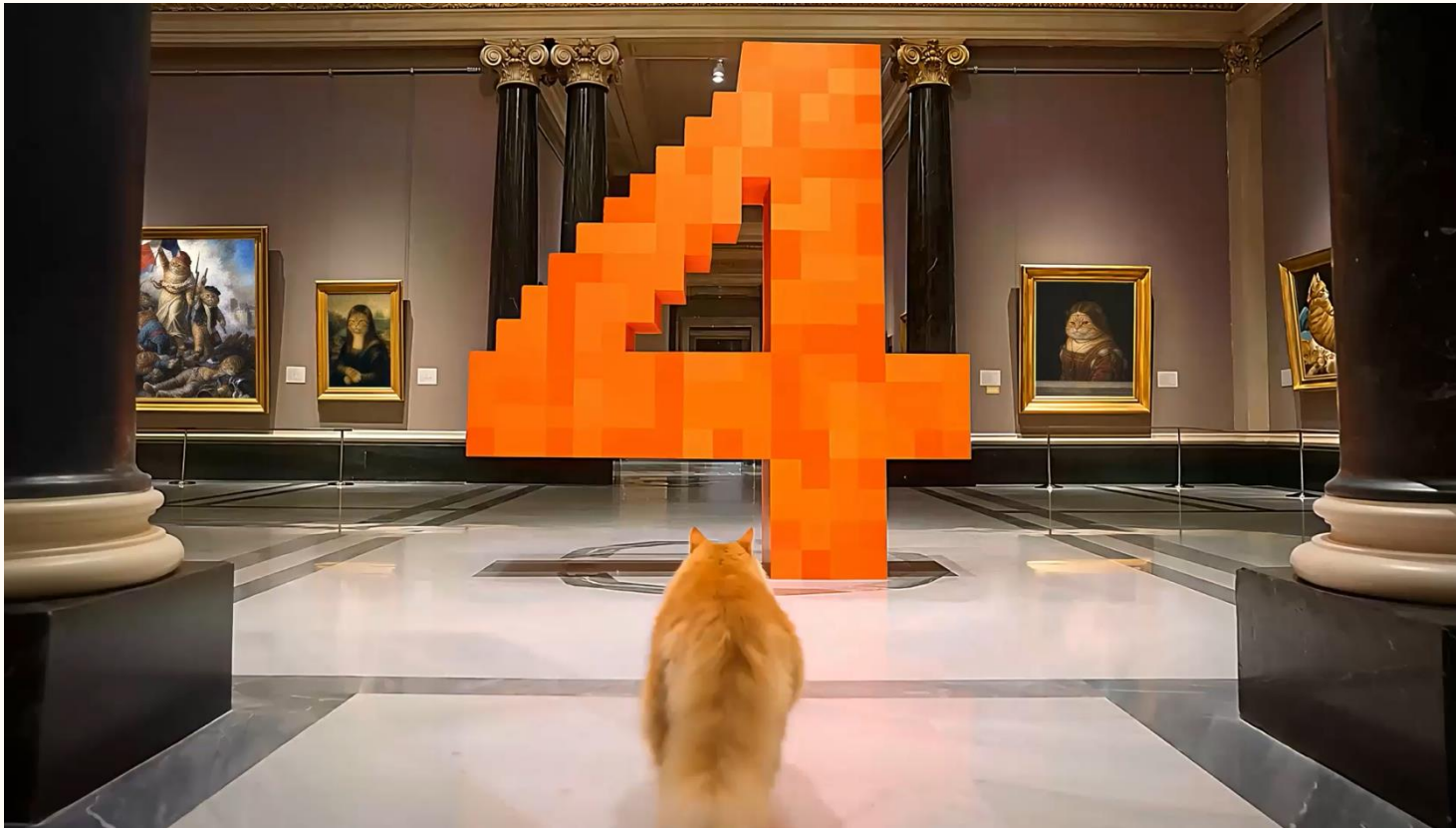
Upper-body centreline



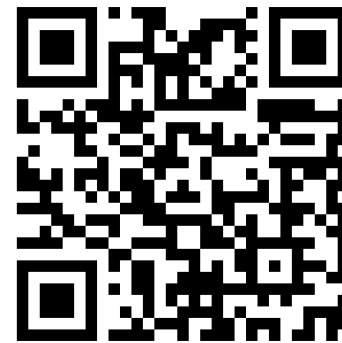
Underbody centreline



Thanks



AB-UPT paper



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Mistral AI



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Special thanks to Tobias Kronlachner who
trained the models